AETERNA SINGULARITY // WET-LAB EXPERIMENTAL TRIAL PROTOCOL

Closed-Loop Bioelectric Epigenetic Rejuvenation & Horvath Clock Validation Suite

Architect: Dimitar Prodromov (ID: 101327948)	Target Hardware: STM32H743ZI Cortex-M7 @ 480 MHz
European Commission PIC: 865986222	Firmware Module: firmware_care_stm32h7_core.rs
EU Coordinator ID: 101327948	Reference DOI: 10.5281/zenodo.23434535
Corporate Entity: AETERNA Technologies EOOD, Pomorie HQ	Compliance: ISO 17025 | EU MDR Class IIb/III

§ 1. TRIAL OBJECTIVE & SCIENTIFIC HYPOTHESIS

Core Scientific Hypothesis: Real-time 10 kHz closed-loop bioelectric field potentials (Robert O. Becker restoration currents) generated by the STM32H7 8D EKF-CARE controller directly guide aged Human Dermal Fibroblasts (HDF, donor age 65–75 years) down Waddington's epigenetic landscape, achieving:

- Significant Horvath Clock Reversal: Reduction in DNA methylation age of ΔAge ≤ -20.0 years ($p < 10^{-5}$).
- Zero Neoplastic Risk: Strict somatic boundary confinement without oncogenic dedifferentiation ($P_{cancer} \leq 10^{-89}$).
- Senescence Clearance: >75% reduction in SA-β-gal positive cells and clearance of p16^INK4a / p21^CIP1 arrest markers.

§ 2. BIOLOGICAL MODEL & CELL CULTURE SYSTEM

Primary Human Dermal Fibroblasts (HDF): Sourced from healthy human donors aged 68 ± 4 years (N = 3 biological donors, male & female, non-smokers, no neoplastic history). Companion replicative senescence cohort tested at passage P14–P16 (PDL > 45).

Environmental Standard: High-glucose DMEM supplemented with 10% dialyzed Fetal Bovine Serum (dFBS, heat-inactivated at 56°C to eliminate microRNA confounding), 1% Pen-Strep, 1 mM sodium pyruvate. Incubated at 37.0°C ± 0.2°C, 5.0% CO_2, 95% relative humidity. Medium electrical conductivity calibrated to $\sigma = 1.45 \pm 0.05$ S/m.

§ 3. PHYSICAL WET-LAB APPARATUS & ELECTROCHEMICAL INTERFACING

To eliminate cytotoxic Faradaic reactions, metal dissolution, and pH gradients, the STM32H7 hardware interfaces with the culture via an engineered non-Faradaic boundary:

- Actuation Stage (DAC): AD5686R (16-bit) buffered by ADA4528 zero-drift op-amps feeding an enhanced Howland current pump. Output is capacitively coupled through a 10 µF tantalum DC-blocking capacitor, ensuring exactly zero net DC drift ($\int I_{stim}\, dt = 0.000$ nA).
- Electrode Metallurgy: Working and counter electrodes comprise sintered Platinum-Iridium (Pt/Ir 90/10) wires with electrochemically platinized surfaces ($Z < 15\ \Omega \cdot cm^2$ at 10 kHz). Maximum current density $J_{stim} \leq 0.45\ \mu A/cm^2$ (water electrolysis barrier: 1.0 µA/cm²).
- Sensing Stage (AFE ADC): Texas Instruments ADS1299 (24-bit, 1.0 µVpp noise floor, PGA gain = 12) coupled via non-polarizable Ag/AgCl reference microelectrodes in 3M KCl agar salt bridges ($Z_{in} > 10$ GΩ).
- Analog Safety Tripwire: STM32H7 internal comparator COMP1 (PA1) hardwired to $V_{trip} = -15.0$ mV. Triggers solid-state optical relays in < 42 ns if abnormal membrane depolarization occurs.

§ 4. EXPERIMENTAL ARMS & COHORT MATRIX

Cohort	Designation	Condition / Regimen	Sample Size	Primary Scientific Role
Cohort A	Sham Control	Electrodes immersed, 0.0 µA stimulation	N = 6	Baseline for spontaneous epigenetic drift
Cohort B	Yamanaka Control	Transient OSKM mRNA transfection (0.5 µg)	N = 6	Positive control for genetic pluripotency
Cohort C	Open-Loop DC	Unregulated static Becker current (-50 mV DC)	N = 6	Demonstrates instability of open-loop DC
Cohort D	AETERNA CARE™	STM32H7 10 kHz Closed-Loop EKF-CARE	N = 6	Safe somatic epigenetic age reversal

§ 5. MOLECULAR ENDPOINTS & INDEPENDENT HORVATH CLOCK ASSAY

• Primary Epigenetic Endpoint: DNA extraction (Qiagen DNeasy), bisulfite conversion (Zymo EZ Gold), and microarray hybridization on Illumina Infinium MethylationEPIC v2.0 BeadChip (>935,000 CpG sites). DNA methylation age calculated via the 353-CpG Horvath Pan-Tissue Clock, 391-CpG Skin & Blood Clock (Horvath 2018), and PhenoAge.

• Target Reversal: ΔAge = DNAmAge(Day 7) - DNAmAge(Baseline) ≤ -20.0 years ($p < 10^{-5}$).

• Teratoma & Dedifferentiation Safety Audit: RT-qPCR and western blot for POU5F1 (OCT4), SOX2, and NANOG. Expression in Cohort D must remain at somatic baseline (< 0.05% of Cohort B Yamanaka levels, $p < 10^{-12}$), formally confirming $P_{cancer} \leq 10^{-89}$.

• Cellular Senescence & Bioenergetics: Cytochemical SA-β-gal assay (>75% reduction), p16^INK4a / p21^CIP1 protein quantification (>60% clearance), and Seahorse XF Cell Mito Stress Test for ATP-coupled respiration and spare respiratory capacity.

§ 6. INDEPENDENT THIRD-PARTY EXECUTION & TIMELINE

To ensure unquestionable regulatory and peer-review integrity, all genomic samples receive 128-bit randomized cryptographic hash IDs (UUIDv4) and are dispatched double-blind to an independent, ISO 17025 certified European genomics facility (e.g. Macrogen Europe / Eurofins Genomics). Total trial duration: 14 weeks across 6 operational milestones (€45,000 budget).

PROTOCOL APPROVED BY SOVEREIGN AUTHORITY:
Dimitar Prodromov, Sole Architect
AETERNA Technologies EOOD, Pomorie HQ
Cryptographic Seal: 0x41_45_54_45_52_4e_41_5f_4c_4f_47_4f_53_5f_44_49_4d_49_54_41_52_5f_50_52_4f_44_52_4f_4d_56_21

INSTITUTIONAL METRICS:
• EC PIC: 865986222
• EU Coordinator ID: 101327948
• CERN / Zenodo: 10.5281/zenodo.23434535
• Status: CERTIFIED READY FOR WET-LAB EXECUTION